An Analysis of Population Structure of Arctic Fox in Greenland and Arctic Circle Area

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Population Genetics Course 2026, Group 4

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Abstract

This study investigates the population structure of the Arctic fox (*Vulpes lagopus*) in Greenland and the circumpolar region, focusing on the role of sea-ice connectivity in shaping gene flow. We analyzed a high-density genomic dataset of 0.8 million SNPs from Greenland, Canada, and Siberia. Genetic structures were characterized using PCA and ADMIXTURE clustering following quality control and LD pruning via PLINK, with evalAdmix ensuring the robustness of ancestral inferences. Genetic differentiation and diversity were further quantified using pairwise Fst and runs of homozygosity (ROH) analysis. The results reveal a distinct geographical structure among Greenlandic populations, with evidence of significant sea-ice mediated connectivity. PCA identified three explicit immigrants in Scoresbysund whose genomic profiles align with distant circumpolar metapopulation. Notably, several individuals from Zackenberg and Kangerlussuaq exhibited a genetic inclination toward the circumpolar metapopulation in the PCA. ADMIXTURE analysis confirmed that these individuals harbor both local and external ancestral components, suggesting they are admixed descendants of historical migrants rather than primary immigrants. These findings demonstrate that sea-ice bridges facilitate not only long-distance individual dispersal but also substantial inter-population gene flow and genetic introgression.

Keywords: Arctic fox; Population structure; Sea-ice connectivity; Gene flow; Genetic admixture

Introduction

The arctic fox (*Vulpes lagopus*) is a small-sized carnivore which is adapted to arctic environments, and is characterized by short morphology features that minimize heat loss. It is a circumpolar species and currently inhabits tundra regions across Greenland, North America, Eurasia, Iceland. Arctic fox populations are relatively abundant, although there are some fluctuations relating to prey availability, particularly lemmings, which represent the species' main food source[1].

Arctic foxes exhibit a remarkable movement ecology, with individuals being able to travel very long distances, across sea ice and terrestrial environments. Satellite telemetry studies have shown that both juveniles and adults disperse across large Arctic areas, often covering total distances of several thousand kilometers. This highlights the species' exceptional dispersal capacity[2]. These long-distance movements are facilitated by the presence of sea ice, which allows Arctic foxes to inhabit distant areas. Sea ice acts as a bridge, connecting far flung tundra habitats and distant populations across the circumpolar distribution of the species. Arctic foxes seasonally move over the sea ice, to find food sources. These movements are mostly connected to lemming population cycles, and can lead them to new areas, influencing the gene flow[3].

Greenland is a unique and important system for studying Arctic fox population structure. This dynamic can be largely attributed to Greenland's unique biogeographical paradigm. While Greenland functions as a strictly isolated landmass separated by expansive marine barriers during the warmer months, the extensive formation of winter sea ice establishes ephemeral terrestrial corridors linking it to both the North American and Eurasian continents. For highly mobile Arctic species, these seasonal cryospheric bridges serve as critical dispersal conduits. They facilitate episodic, long-distance migration events

across the Holarctic region, thereby enabling periodic gene flow. Such intermittent genetic admixture plays a pivotal evolutionary role, effectively mitigating the loss of heterozygosity and counteracting the effects of genetic drift that would otherwise dominate in geographically restricted insular populations[3,4]. This combination of isolation and connectivity allows both regional differentiation and population mixing, making Greenland a very unique and ideal study area for understanding the interplay between movement ecology and genetic structure.

Genetic studies have shown that populations connected by sea ice generally have low genetic differentiation, indicating extensive gene flow. In contrast, populations isolated from sea ice, such as those in Iceland, and parts of Scandinavia, exhibit higher genetic divergence. These findings demonstrate the important role of sea ice connectivity in creating population genetic structures[3,4]. Mitochondrial analyses also revealed that many haplotypes are shared across distant, ice-connected regions, including North America, northern Greenland, Svalbard, Siberia, reflecting an extensive historical gene flow[5]. Recent microsatellite studies further showed that even among connected populations, small local differences exist because related individuals tend to stay close together[6]. In addition, Arctic foxes can be divided into two ecotypes, lemming and coastal, based on their primary food sources. This food preference leads to habitat imprinting, which means that foxes prefer to inhabit environments similar to their natal habitat, creating genetic subdivisions even without physical barriers[7]. Together, these studies show the importance of interactions among movement ecology, sea-ice connectivity, and historical population processes in shaping Arctic foxes' genetic structure.

In this study, our goal is to investigate the populational structure of arctic foxes in Greenland, focusing on how movement patterns and sea-ice connectivity have influenced gene flow among populations. We analyzed samples from Zackenberg(n=5), Scoresbysund(n=12), Qanisartuut(n=11), and Kangerlussuaq(n=10), and included also samples from Canada and Siberia(n=9) which contain ~0.8 million bi-allelic SNPs to understand both local and regional patterns of connectivity (Fig 1).

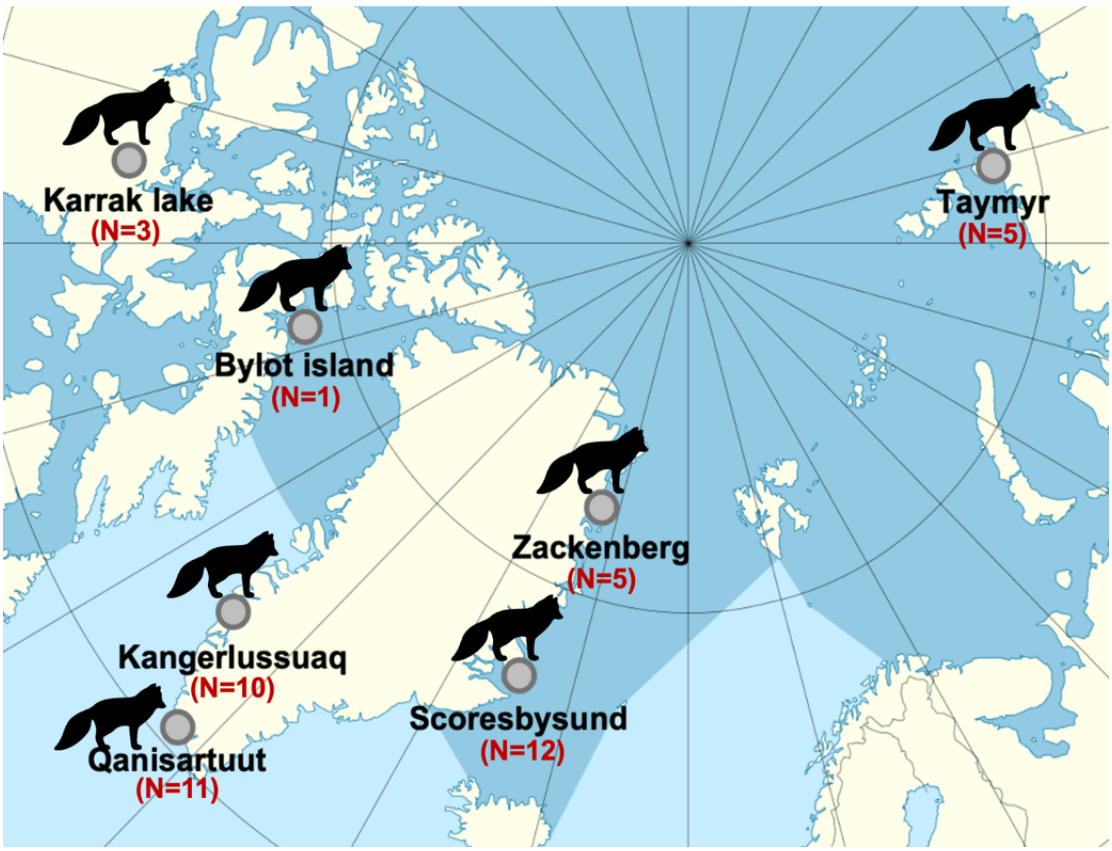

Fig 1. Geographical distribution of the study sites across the Arctic region. This map illustrates the locations and samples numbers of the seven regions in this study: Karrak Lake, Bylot Island, Kangerlussuaq, Qanisartuut, Scoresbysund, Zackenberg, and Taymyr.

Methods

Study Species and Genomic Dataset

The genomic data in this study consisted of an imputed and thinned SNP dataset of the Arctic fox (*Vulpes lagopus*), provided by the Population Genetics Course at the University of Copenhagen. To ensure computational efficiency while maintaining genome-wide representation, the original dataset was subsampled to a high-quality subset of markers. These sample data are sampled from Greenland (Kangerlussuaq, Qanisartuut, Scoresbysund, Zackenberg), Canada (Bylot island, Karrak lake) and Siberia (Taymyr).

Genotype Quality Control and PCA

Genotype quality control and Principal Component Analysis (PCA) were performed using PLINK v1.9[8]. SNPs were filtered based on missingness (--geno 0) and minor allele frequency (--maf 0.05) during the initial data processing stage. To ensure marker independence and minimize redundancy, linkage disequilibrium (LD) pruning was applied ($r^2 < 0.1$, window size 50, step 10). The resulting set of independent SNPs was then used for PCA to characterize the population structure.

Population Structure and Admixture Analysis

Population genetic structure was inferred using ADMIXTURE (v1.3.0)[9] based on the pruned SNP dataset. The optimal number of ancestral clusters was evaluated by testing K values ranging from 2 to 7. Stability and convergence of the clustering were ensured by conducting 100 independent iterations for each K using distinct random seeds. The resulting ancestry proportions (Q-matrices) and allele frequencies (P-matrices) were recorded for each run to facilitate downstream cross-validation and cluster alignment. For each tested K, the run with the highest log-likelihood was selected. The evalAdmix (v0.95)[10] was employed to evaluate inter-individual residual correlations, ensuring the reliability of the inferred population structure. A high correlation of residuals indicates a potential underfitting or over-specification of K. This step ensured that the chosen ancestral clusters accurately represented the underlying genetic variation without significant systematic bias.

Genetic Differentiation

Genetic differentiation between geographical regions was quantified using pairwise F_{ST} estimators. Calculations were performed following the Weir and Cockerham (1984) method[11], implemented via a custom R script provided in exercise in our course. This approach accounts for varying sample sizes and provides a weighted average across all thinned, non-missing SNP loci. All possible pairwise comparisons between identified regions were conducted, and the resulting F_{ST} values were used to assess the degree of genetic isolation among Arctic fox populations.

Heterozygosity and Runs of Homozygosity

Individual observed heterozygosity and runs of homozygosity (ROH) were aggregated by region. Non-parametric methods were employed due to the non-normal nature of the data: the Kruskal–Wallis test was used for global comparisons among regions, followed by Wilcoxon rank-sum tests for pairwise comparisons. All P-values were corrected for multiple testing.

Results

Principal Component Analysis (PCA) revealed a distinct genetic structure among the four Greenlandic Arctic fox populations, with most individuals forming region-specific clusters. However, distant groups from Canada and Siberia tend to cluster. Notably, three individuals (ArcFoxSamp_128, 144, 152) from the Scoresbysund region (colored in orange) did not cluster with their geographical neighbors. Instead, these outliers exhibited strong genetic similarity with extrinsic populations from Canada and Siberia, positioning them far from the main Greenlandic cluster and within the genetic space defined by the circumpolar metapopulation.

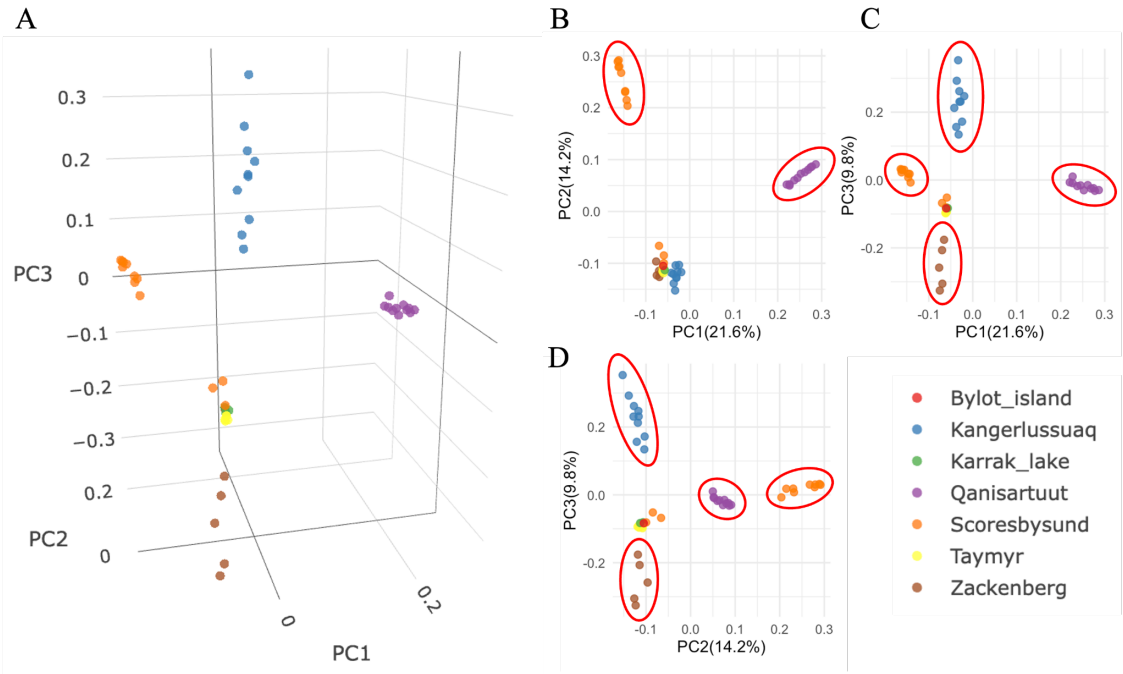

Fig 2. Principal Component Analysis (PCA) of Arctic fox populations. Scatter plots show the first three principal components (PCs) explaining the major genetic variation across individuals. (Left: PC1 vs PC2; Center: PC1 vs PC3; Right: PC2 vs PC3). Individuals are color-coded by sampling region. Red circle indicates these potential clusters of subpopulations.

To determine the most representative number of ancestral clusters (K), we performed ADMIXTURE analysis for K=2 to 7, with 100 independent iterations per K to ensure global optimization. For each K value, the iteration with the highest log-likelihood was identified as the best-fit model: specifically, Run 27 for K=2, Run 68 for K=3, Run 93 for K=4, Run 63 for K=5, Run 25 for K=6, and Run 78 for K=7 .The robustness of these top-ranking models was further validated using evalAdmix. The analysis of residual correlations between individuals revealed that K=5 provided the most stable and biologically parsimonious fit to our data, with the lowest systematic bias compared to other values. Consequently, the results from K=5 (Run 63) were selected for the final characterization of the Arctic fox population structure, as this model best captured the genetic differentiation between the Greenlandic regions and the circumpolar metapopulation.

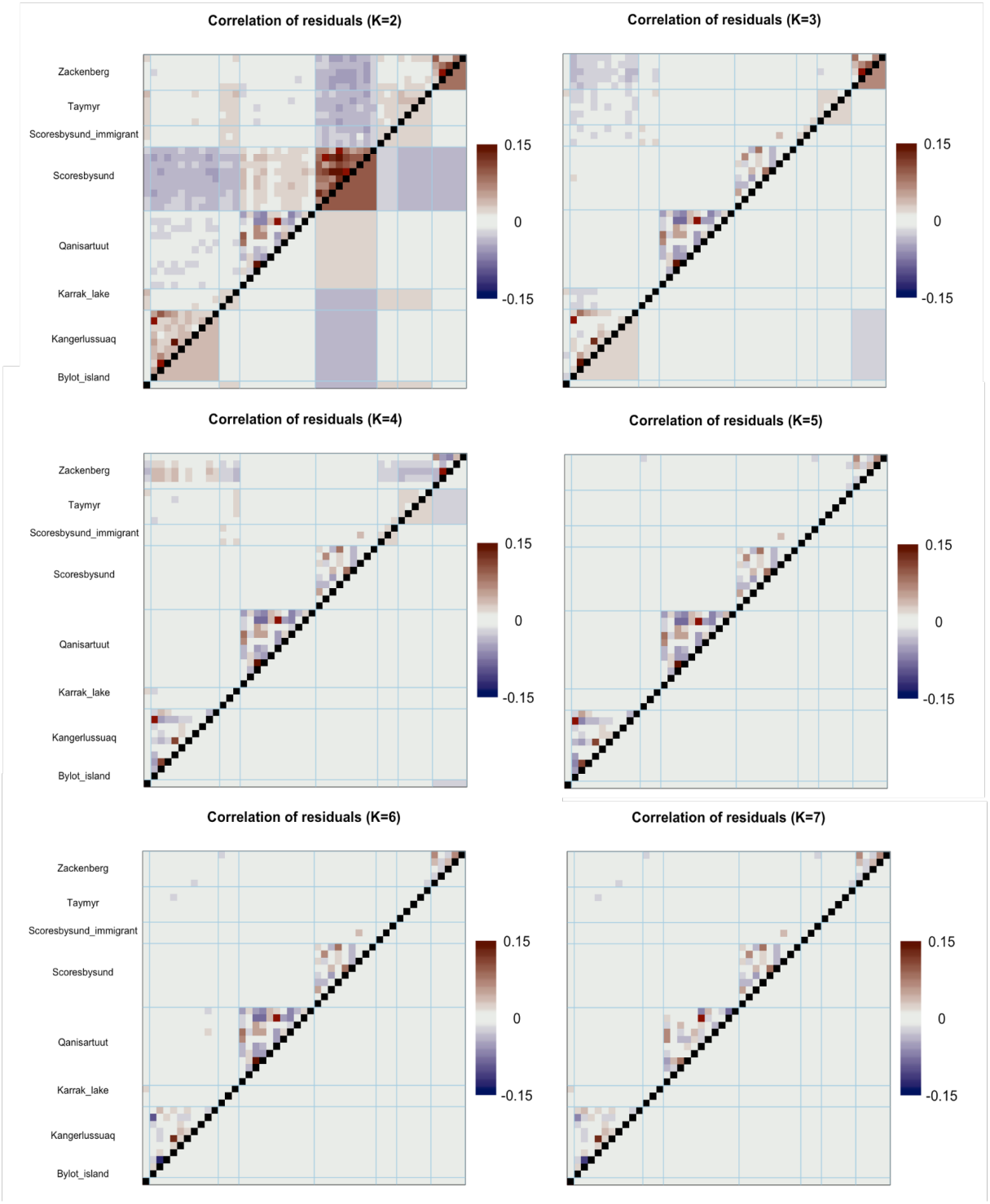

Fig 3. Evaluation of ADMIXTURE model fit using residual correlation matrices for K=2 to 7. The heatmaps display the correlation of residuals between individuals across different K values. Darker colors (brown/blue) indicate systematic deviations where the model fails to explain the genetic data (under-fitting). At K=5, the matrix shows a predominantly uniform light-grey distribution, indicating that the five-cluster model effectively captures the underlying population structure with minimal residual correlation. This validates K=5 as the most parsimonious and robust model for the dataset.

The ADMIXTURE analysis at K=5, identified as the most parsimonious model via evalAdmix residual correlations, revealed a pronounced genetic structure among Greenlandic Arctic fox populations. The four primary regions Qanisartuut (QAN), Scoresbysund (SCO), Zackenberg (ZAC), and Kangerlussuaq (KAN) exhibited high levels of genetic independence, with most individuals assigned to a single, region-specific ancestral component. This suggests early divergence and minimal recent gene flow between these established populations. However, striking genetic outliers were identified within the

northern groups. Specifically, three individuals from Scoresbysund (SCIM) displayed an ancestry profile dominated by a "green" component, indicating high genetic affinity with Canadian (Bylot Island and Karrak Lake) and Siberian (Taymyr) metapopulation rather than their geographical neighbors. Similarly, individuals (ArcFoxSamp_104, 183, and 191) in the Zackenberg population exhibited substantial green and blue components, suggesting they represent non-local lineages. While the Kangerlussuaq population also displayed an admixed profile — characterized by a dominant red component alongside a persistent green ancestral signature (ArcFoxSamp_149, 165, 173, and 189) — this subtle admixture was less discernible in the PCA. Conversely, the ADMIXTURE results for Zackenberg are explicitly corroborated by the PCA, where several Zackenberg individuals (represented by brown data points) cluster in close proximity to the circumpolar metapopulation further validating their status as potential historical migrants.

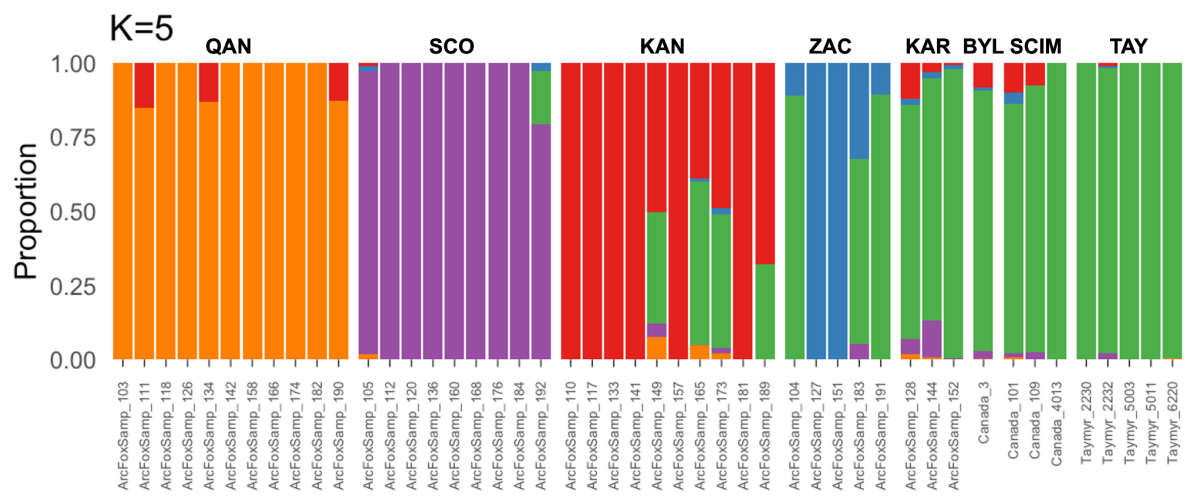

Fig 4. Genetic structure of Arctic fox populations inferred by ADMIXTURE at K=5. Each vertical bar represents an individual, and the colors represent the proportional assignment to five ancestral clusters. Individuals are grouped by geographic region: QAN: Qanisartuut; SCO: Scoresbysund; KAN: Kangerlussuaq; ZAC: Zackenberg; KAR: Karrak Lake; BYL: Bylot Island; SCIM: Scoresbysund Immigrants; TAY: Taymyr.

The pairwise F_{ST} analysis, calculated using the Weir and Cockerham (1984) estimator, reveals a complex population structure for the Arctic fox, characterized by both high connectivity across vast distances and localized genetic isolation. A striking feature of the genomic landscape is the near-zero differentiation ($F_{ST} \approx 0.000 - 0.001$) between the Siberian (Taymyr) and Canadian (Bylot Island, Karrak Lake) populations. Despite the thousands of kilometers separating these regions, the negligible F_{ST} values suggest an extremely high level of gene flow, likely facilitated by the species' use of seasonal sea ice as a corridor for long-distance dispersal. In contrast, populations in Greenland exhibit significantly higher levels of genetic structure; Qanisartuut, in particular, shows the highest degree of isolation with F_{ST} values ranging from 0.087 to 0.141 against all other groups. This isolation is probably driven by the more sedentary "coastal" ecotype of foxes in Southern Greenland, which rely less on sea-ice-driven migrations than their high-arctic counterparts. Furthermore, the inclusion of the "Scoresbysund_immigrant" provides a compelling case study for real-time gene flow; although sampled in Scoresbysund, this individual is genetically indistinguishable from the Taymyr and Canadian populations, highlighting the stark contrast between resident lineages and transient migrants. These relationships are visually summarized by the hierarchical clustering dendrogram, which bifurcates the study sites into a divergent Greenlandic cluster and a highly mobile, panmictic-like circumpolar cluster.

Table 1. Summary of Genetic Differentiation Patterns

Comparison	F_{ST} Range	Interpretation
Taymyr vs. Canada	0.000 – 0.001	High Gene Flow / Panmixia
Qanisartuut vs. All	0.087 – 0.141	Significant Genetic Isolation
Scoresbysund vs. Immigrant	0.059	Resident/Migrant Distinction

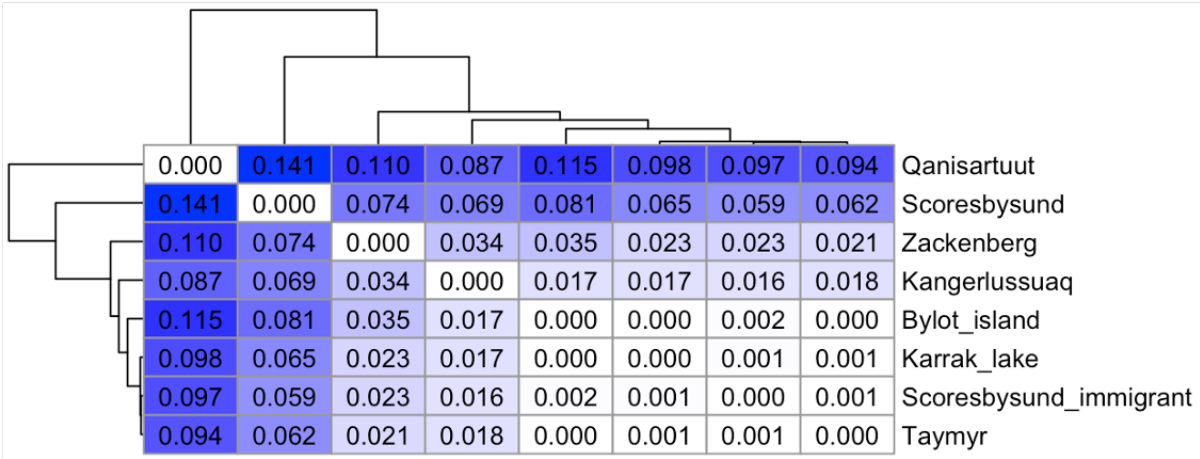

Fig 5. F_{ST} results among Arctic fox populations. A heatmap of pairwise F_{ST} values. The color intensity represents the degree of genetic differentiation, with darker blue indicating higher F_{ST} values and greater genetic isolation. The dendrogram (top and left) illustrates the hierarchical clustering of populations based on genetic similarity.

The mean observed heterozygosity varied among different regions (Fig 6). Qanisartuut had the lowest mean H_0 and Canada and Taymyr had the highest values. Within the different regions, there was variation, particularly in Qanisartuut, likely influenced by individuals with lower heterozygosity values.

Total runs of homozygosity (ROH) length also varied among regions (Fig 7). Individuals from Qanisartuut showed the highest total ROH length, followed by Scoresbysund and Kangerlussuaq, while Canada and Taymyr showed the lowest ROH values. Zackenberg and Scoresbysund_immigrant showed relatively low ROH compared to other Greenland regions. Overall, ROH length differed among regions, indicating differences in levels of homozygosity among Arctic fox populations.

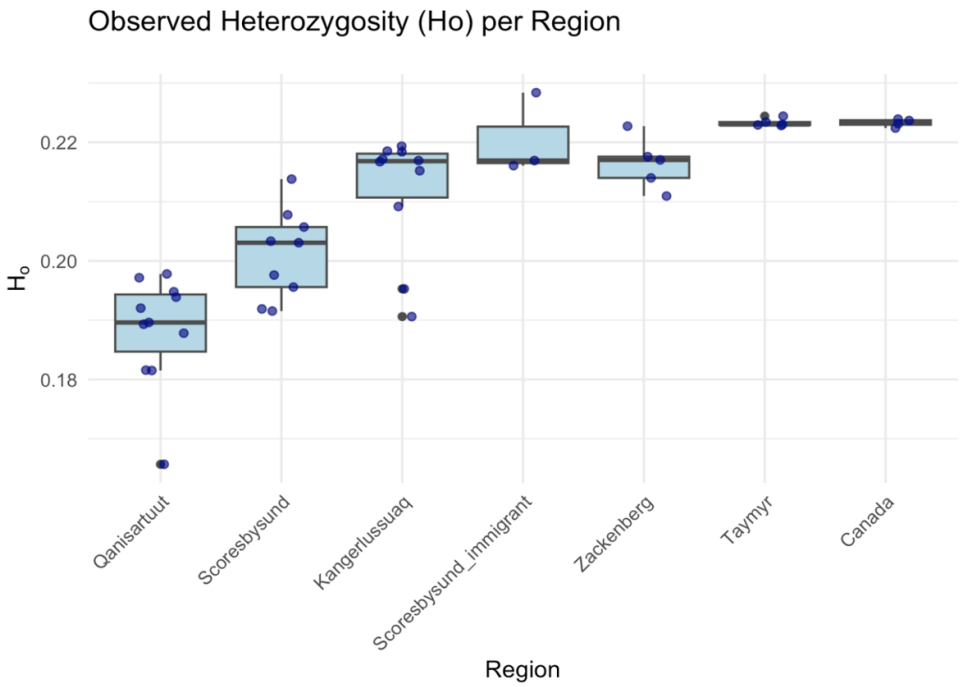

Fig 6. Observed heterozygosity (Ho) per region. Each point represents an individual, and boxplots summarize heterozygosity distribution within each region.

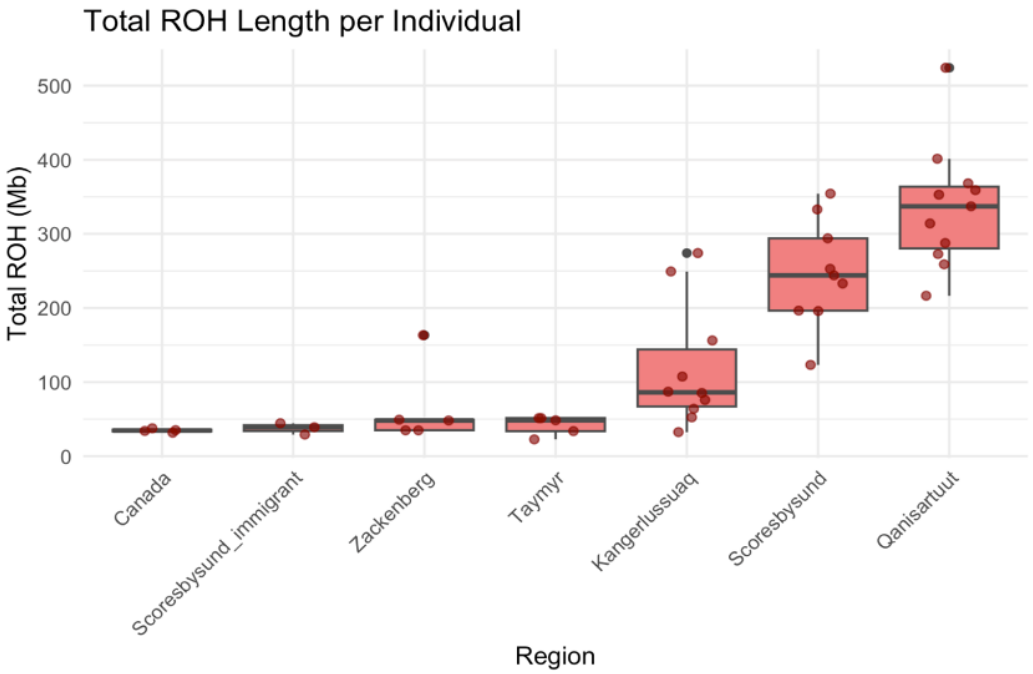

Fig 7. Total runs of homozygosity (ROH) length per individual across regions. Each point represents an individual, and boxplots summarize the distribution of ROH length within each region.

Discussion

In the PCA results, with the exception of the group marked in purple, samples across all projection dimensions of the four 3D PCA plots exhibit dense clustering. Conversely, sample points from the three sampling sites in Greenland show an overall trend of radiating outward from the core distribution area. From the perspective of genetic distance, the Kangerlussuaq group (blue) and the Zackenberg group (brown) display distinct genetic characteristics, warranting significant research attention. In the projection plane defined by PC1 and PC2, Kangerlussuaq and Zackenberg samples are highly clustered and positioned close to the metapopulation (Canada and Siberia); in the other two projection dimensions, certain samples within the Zackenberg group exhibit similarities to the metapopulation.

Further investigation integrated with ADMIXTURE ancestry profiles reveals that three samples in the Zackenberg group harbor a substantial proportion of the green ancestral component while lacking the red component, suggesting a close genetic affinity with the Taymyr group. In preliminary analyses, immigration signals within the Scoresbysund group were identified via PCA; consequently, these individuals were analyzed separately in the ADMIXTURE stage and designated as Scoresbysund_immigrant. In contrast, the immigration signals present in the Kangerlussuaq and Zackenberg groups were not immediately noticed during the PCA phase, as their associated admixture events are not recent but represent historical gene flow. Specifically, the Zackenberg group retains a large proportion of the green ancestral component, indicating a relatively more recent immigration event. For the Kangerlussuaq group, the green component is significantly reduced, with the red component dominating the ancestry, suggesting a more ancient timing for its corresponding immigration. Geographically, Kangerlussuaq is situated near the Scoresbysund; the presence of a minor orange ancestral component in these samples aligns with both the geographical distribution and the migration patterns towards the Canadian region.

The limited sample size of the Zackenberg group (n=5) introduces potential bias. Specifically, the observed 2:3 ratio of blue to green ancestry components obscures the identification of the true "native" lineage. While we tentatively assign blue as the resident component, the high proportion of exogenous green ancestry necessitates a larger dataset to confirm these admixture dynamics.

In summary, the immigrant individuals detected in the Scoresbysund group represent recent migrations; these samples possess almost none of the genetic signatures characteristic of the Scoresbysund group, yet were found within the Scoresbysund region. In contrast, the four samples in the Kangerlussuaq group and the three samples in the Zackenberg group are not direct immigrants themselves. Instead, they are more likely admixed descendants resulting from genetic exchange between immigrants and the resident population, thus harboring genetic traits from local, Canadian, and Siberian populations simultaneously.

Previous microsatellite and mtDNA studies have reported significant genetic differentiation among Greenlandic Arctic fox localities such as Scoresbysund, Qaqortoq, and Kangerlussuaq[12]. This is consistent with our SNP-based evidence of strong structure in Greenland, particularly the distinctiveness of South Greenland (Qanisartuut) and moderate differentiation at Scoresbysund.

Observed heterozygosity and runs of homozygosity analyses showed variation among populations. Because our dataset includes only 47 individuals in total, the results might be affected. Qanisartuut showed the lowest heterozygosity and the highest total ROH, including some clear outliers. This could be consistent with reduced diversity, but we cannot make strong conclusions about a bottleneck or founder effect from these measures alone.

We also tested for a bottleneck or founder effect in Greenlandic Arctic foxes using π, Watterson's θ, Tajima's D, and the folded SFS (Fig 8, A-D). Although Tajima's D is negative in all populations, the values are extremely negative and likely reflect artifacts of imputation, SNP thinning, and small sample size rather than clean demographic signal. The folded SFS is also low-resolution because downsampling is constrained by the smallest sample, and folding further reduces interpretability. Diversity statistics do not show a strong, consistent reduction in Greenland relative to source populations. Therefore, our summary statistics do not reveal a clear bottleneck or founder-effect signal in Greenland, but the data are not well suited for robust demographic inference, so a bottleneck cannot be excluded.

Several limitations of this study warrant consideration. Notably, the stringent data filtering applied may have obscured subtle signals necessary for deeper analyses of population history and gene flow. Future research should integrate datasets from a broader range of sources to enable higher-resolution reconstructions of the origins and temporal dynamics of the identified migration events. Furthermore, quantifying the genetic divergence between the Qanisartuut and Meta populations could facilitate the estimation of historical ice bridge formation periods, thereby offering novel insights into paleoclimatology and biogeography.

Reference

1. Fuglei, E. & Ims, R. A. Global warming and effects on the Arctic fox. Sci. Prog. 91, 175–191 (2008).

2. Gravel, R., Lai, S. & Berteaux, D. Long-term satellite tracking reveals patterns of long-distance dispersal in juvenile and adult Arctic foxes (Vulpes lagopus). R. Soc. Open Sci. 10, 220729 (2023).

3. Norén, K. et al. Arctic fox Vulpes lagopus population structure: circumpolar patterns and processes. Oikos 120, 873–885 (2011).

4. Norén, K. et al. Pulses of movement across the sea ice: population connectivity and temporal genetic structure in the arctic fox. Oecologia 166, 973–984 (2011).

5. DALÉN, L. et al. Population history and genetic structure of a circumpolar species: the arctic fox. Biol. J. Linn. Soc. 84, 79–89 (2005).

6. Lai, S., Quiles, A., Lambourdière, J., Berteaux, D. & Lalis, A. Fine-scale population genetic structure of arctic foxes (Vulpes lagopus) in the High Arctic. BMC Res. Notes 10, 663 (2017).

7. Eide, N. E., Stien, A., Prestrud, P., Yoccoz, N. G. & Fuglei, E. Reproductive responses to spatial and temporal prey availability in a coastal Arctic fox population. J. Anim. Ecol. 81, 640–648 (2012).

8. Purcell, S. et al. PLINK: A Tool Set for Whole-Genome Association and Population-Based Linkage Analyses. Am. J. Hum. Genet. 81, 559–575 (2007).

9. Alexander, D. H., Novembre, J. & Lange, K. Fast model-based estimation of ancestry in unrelated individuals. Genome Res. 19, 1655–1664 (2009).

10. Garcia-Erill, G. & Albrechtsen, A. Evaluation of model fit of inferred admixture proportions. Mol. Ecol. Resour. 20, 936–949 (2020).

11. Weir, B. S. & Cockerham, C. C. ESTIMATING F-STATISTICS FOR THE ANALYSIS OF POPULATION STRUCTURE. Evol. Int. J. Org. Evol. 38, 1358–1370 (1984).

12. Meinke, P. G., Kapel, C. M. O. & Arctander, P. Genetic differentiation of populations of Greenlandic Arctic fox. Polar Res. 20, 75–83 (2001).

Supplementary Information

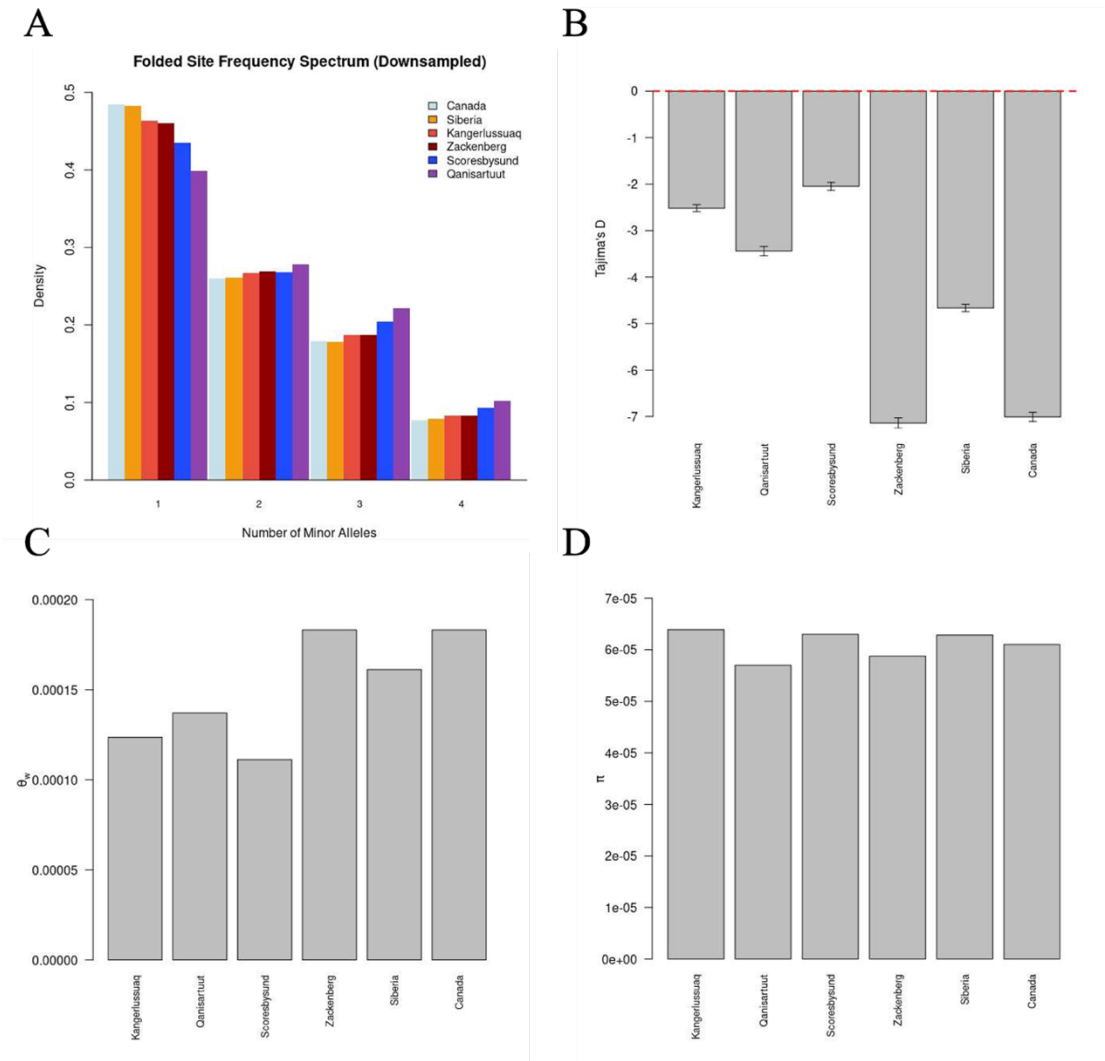

Fig 8. Barplot results of Folded SFS, Tajima'D, Watterson's θ, and π

Declaration of using generative AI tools (for students)

☑ I/we have used generative AI as an aid/tool (please tick)
☐ I/we have NOT used generative AI as an aid/tool (please tick)

List which GAI tools you have used and include the link to the platform:
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2) Gemini: https://gemini.google.com/app
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Describe how generative AI has been used in the exam paper:
1) Purpose

   Grammar: To correct English grammar issues within the text.

   Content Organization: To help properly structure the content of the report.

   Code: To aid in both understanding and writing code.
2) Work phase

   Drafting: Used while writing the final report.

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3) What did you do with the output?

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